KMIT KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Narayanaguda, Hyderabad.

B.TECH III YEAR II SEM (KR23) –I-MID EXAMINATIONS, FEBRUARY/MARCH - 2026

SUBJECT: MIDDLEWARE TECHNOLOGIES **PART-B** **DATE: 25/2/2026-AN**

BRANCH: CSE Duration: 1Hr 40 Min. **Max.Marks:20 M**

ANSWER ANY FOUR QUESTIONS **4*5=20M**

Q.No	Questions	Marks	CO	BL
1.	A real-time food delivery platform QuickBite uses Redis to manage its operations. Due to system design constraints, the architect has specified strict requirements for how data must be stored and accessed. You must follow these mandatory technical constraints: 1. The system must store restaurants "Dominos", "KFC", and "BurgerKing") in the exact order of registration, and duplicate restaurant names must be allowed. 2. The order IDs 101, 102, and 103 processing system must follow the First-In-First-Out (FIFO) principle, where the first order placed is processed first, and the order must be removed from the system after processing. 3. Customer information (name, phone, city) must be stored such that individual fields can be retrieved, updated, or modified independently without affecting other fields. (ID: 1001, Name: Rahul, Phone: 9876543210, City: Hyderabad). 4. The system must maintain the number of orders per restaurant and support atomic increment operations, ensuring accurate counting without overwriting existing values. (order count of "Dominos" by 1). 5. The system must store restaurant ratings (Dominos (4.5), KFC (4.2), BurgerKing (4.7)) and automatically maintain ranking based on rating scores, allowing efficient retrieval of the highest-rated restaurant. Based on the above strict constraints, write the appropriate Redis CLI commands to perform the operations.	5	2	3
2. a)	Explain the network partition problem (split-brain scenario) in distributed systems and describe the techniques used to achieve fault tolerance.	3	1	2
2. b)	Explain the spectrum of consistency models used in distributed systems with examples.	2	1	2
3. a)	Explain the architecture and components of a modern cloud-native middleware stack.	3	1	3
3. b)	What is Messaging Middleware (Message-Oriented Middleware)? Mention any two of its key functions.	2	1	2

4.	An international airport called SkyTrack uses Redis to manage baggage handling, passenger baggage details, security clearance, baggage priority handling, and baggage counters. Due to system design constraints, the architect has specified strict requirements for how data must be stored and accessed. You must follow these mandatory technical constraints: 1. The system must store baggage IDs B101, B102, and B104 in the exact sequence of loading into the aircraft. The system must allow insertion of baggage before or after a specific baggage ID and must allow modification of baggage at a specific position. The system must insert baggage B103 after B102 and modify baggage ID B104 to B105. 2. Baggage information must be stored such that individual attributes (owner_name, flight_no, destination, weight) can be accessed independently. The system must prevent overwriting of the owner_name field if it already exists and must verify whether baggage B101 exists before storing its details (Owner: Rahul, Flight: AI202, Destination: Dubai, Weight: 23kg). 3. The system must maintain two collections: security_pending and security_cleared, where duplicate baggage IDs are not allowed. The system must support fast membership checking and must allow movement of baggage B101 from security_pending to security_cleared. 4. The system must store baggage priority scores (B101: 5, B102: 3, B103: 8, B105: 6) and automatically maintain ranking based on priority. The system must support retrieving the rank of baggage B103, retrieving baggage within a priority score range, and removing baggage after loading. 5. The system must maintain a counter for total baggage loaded. The counter must support atomic increment, atomic decrement, increment by a specific value, and retrieval of the current value. The system must increment the counter when baggage B101 and B102 are loaded, increment by 3 when baggage B103, B104, B105 are loaded together, decrement when baggage B102 is removed, and retrieve the final counter value.	5	2	3
5. a)	A social media platform frequently retrieves user profile information from a database. Engineers observe that repeated database access increases response time and server load. What Redis-based mechanism can be used to reduce database load and improve response time? Explain in detail.	3	2	3
5. b)	Explain with snippet code, how the system can implement navigation between sections without reloading the entire page.	2	1	2
6. a)	An in-memory data system provides extremely fast performance but engineers notice that some recent updates are lost after a server restart. To prevent this, they introduce mechanisms that periodically save snapshots and also log every update operation to disk. Explain how these mechanisms help prevent data loss.	3	2	2
6. b)	In a React application, multiple nested components need access to shared course information without passing props through intermediate components. Which React Hook is used to access shared data and what problem does it solve. Explain with Snippet code.	2	1	2

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III- B.Tech II Semester – KR23-MID-I Examinations, February/March - 2026

Subject: INTERNET OF THINGS
Branch: CSE

PART-B
Duration: 100 Min.

Date: 26/2/2026-FN
Max. Marks: 20

Answer any FOUR from the following Questions 4X5=20M

Q.No.	Questions	Marks	CO	BL
1	Explain the Physical Design of IOT with suitable diagrams	5	1	2
2.a)	How would APIs help in managing communication between the mobile app and the IOT devices. Classify different types	2	1	2
b)	How does the IOT system functional blocks ensure that the data collected from the wearable devices is securely transmitted to the cloud	3	1	3
3	Give a brief note on IOT communication models with neat diagrams.	5	1	2
4 a)	How does the transducers impact decision-making in IOT systems	2	2	3
b)	Explain the classification of Sensor based on different types.	3	2	2
5 a)	How can the data from the DHT-22 sensor be used to send real-time alerts if the temperature or humidity exceeds predefined thresholds	2	2	2
b)	List out key considerations when selecting an actuator for an IOT application.	3	2	1
6 a)	What types of actuators are commonly used in robotic systems for industrial automation	2	2	1
b)	Explain the working and features of LED in 7-segment display with relevant use cases as examples.	3	2	2

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III- B.Tech II Semester [KR23] , MID- I EXAMINATIONS, February/March-2026

Subject: -INTELLECTUAL PROPERTY RIGHTS Date: 26/2/2026-AN

Branch : CSE Duration: 2Hrs Max. Marks:100

Answer any FIVE from the following Questions **(5*20M)**

Q.No.	Questions	Marks	CO	BL
1	Explain Patents and Trade secrets with examples.	20	CO1	BL2
2	Explain in detail about the international organizations, agencies and treaties.	20	CO1	BL2
3	Define Trade mark and Explain Protectable Trademarks.	20	CO2	BL1/ BL2
4	Discuss Various Laws and treaties governing Trade marks.	20	CO2	BL2
5	What is copyright? Write basic requirements for copy right ability.	20	CO3	BL1
6	Examine different types of works that can not be protected under copy right Act.	20	CO1	BL4
7	Explain the Trade mark registration process.	20	CO2	BL2
8	Explain works that can be protected under copyright Act.	20	CO2	BL2

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III- B.Tech II Semester – KR23-MID-I Examinations, February/March - 2026

Subject: Cloud Computing | PART-B | Date: 24/02/26-FN

Branch: CSE | Duration: 1Hr 40 Min. | Max. Marks: 20

Answer any FOUR from the following Questions **4*5=20M**

Q.No.	Questions	Marks	CO	BL
1.a)	What is block storage and object storage? Differentiate between block storage, object storage based on structure, access method, and use cases.	3	CO2	BL-1
b)	Explain Flynn's Taxonomy and its purpose in classifying computer architectures.	2	CO1	BL-2
2.	What is a VPC and why is it needed in AWS? How would you design a VPC with a public subnet to launch an EC2 instance with internet access, configure security groups for web traffic, and mention the AWS service categories used in this ?	5	CO2	BL-3
3. a)	What kind of AWS service is Amazon S3 and why do we need it? How would you configure Cross-Region Replication (CRR) between S3 buckets, and why are only newly uploaded objects replicated after enabling CRR while older objects are not?	3	CO2	BL-3
b)	What is Bio-Computing? Explain its concept and importance in modern computing and life sciences.	2	CO1	BL-2
4.	Explain the 5-4-3 principles of NIST in cloud computing, highlighting key features, deployment types, and service offerings.	5	CO2	BL-2
5.a)	What is Nano Computing? Explain in detail about various aspects and challenges of Nano Computing.	2	CO1	BL2
b)	What is horizontal and vertical scaling in cloud computing, and under which type of scaling Amazon EBS fits? How does EBS support rapid elasticity-explain it, and what kind of storage service is it?	3	CO2	BL2
6.a)	Why is quantum computing considered a breakthrough technology for solving complex problems when compared to traditional computing? Explain with suitable reasoning	2	CO1	BL2
b)	What Kind of S3 Storage is ? Under which AWS service category does Amazon S3 fall? How would you create an S3 bucket and upload an object, and if the object is not accessible from a browser, what permission settings would you check to make it publicly accessible?	3	CO2	BL2

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B.TECH. III YEAR II SEM (KR23) –I-MID EXAMINATIONS, February/March - 2026

SUBJECT: Competitive Programming | PART-B | DATE: 24/2/2026-AN

BRANCH: CSE | Duration: 1Hr 40 Min. | Max.Marks:20 M

Answer any FOUR from the following Questions $4*5=20M$

Q.No	Questions	Marks	CO	BL
1.a)	What are the advantages sliding window Technique?	1	1	1
1.b)	Explain the brute force approach to solve the Counting Bits problem using bit manipulation techniques. Implement the solution in Java and analyze its time complexity.	4	2	3
2.a)	Explain Bit wise AND, OR and XOR with Examples.	1	2	2
2.b)	Explain the Approach to solve the Diet Plan performance Application using sliding window Technique and Implement the solution in Java and analyze the time complexity.	4	1	4
3.a)	Explain Bit wise Left Shift Operator with Example.	1	2	2
3.b)	Given two sorted arrays and a number x, find the pair whose sum is closest to x and the pair has an element from each array. We are given two arrays ar1[0…m-1] and ar2[0..n-1] and a number x, we need to find the pair ar1[i] + ar2[j] such that absolute value of (ar1[i] + ar2[j] – x) is minimum. Example: Input: ar1[] = {1, 4, 5, 7}; ar2[] = {10, 20, 30, 40}; x = 32 Output: 1 and 30 For the Above Problem Statement, Describe the Two pointer Approach and implement the solution in Java and analyze the time complexity.	4	1	4
4.a)	Why Two pointer approach? How to Recognizing a Two-Pointer Problem.	1	1	2

[P.T.O]

4.b)	We are given an m x n binary matrix grid.In one operation, you can choose any row or column and flip each value in that row or column (i.e., changing all 0's to 1's, and all 1's to 0's).Return true if it is possible to remove all 1's from the grid using any number of operations or false otherwise. Example: Input: grid = [[0,1,0],[1,0,1],[0,1,0]] Output: true Explanation: One possible way to remove all 1's from grid is to: - Flip the middle row & Flip the middle column For the Above Problem Statement, Describe the Approach using Bit manipulation and implement the solution in Java and analyze the time complexity.	4	2	4
5.a)	What is the difference between fixed-size and variable-size sliding windows?	1	1	2
5.b)	Explain the Procedure to Find the Two Non-Repeated Elements in an array using Bit manipulation Technique and write a java Program for the given Application.	4	2	2
6	The given an array of integers nums, there is a sliding window of size k which is moving from the very left of the array to the very right. You can only see the k numbers in the window. Each time the sliding window moves right by one position.Return the max sliding window. Example: Input: nums = [1,3,-1,-3,5,3,6,7], k = 3 Output: [3,3,5,5,6,7] Explanation: Window position Max --------------- ----- [1 3 -1] -3 5 3 6 7 3 1 [3 -1 -3] 5 3 6 7 3 1 3 [-1 -3 5] 3 6 7 5 1 3 -1 [-3 5 3] 6 7 5 1 3 -1 -3 [5 3 6] 7 6 1 3 -1 -3 5 [3 6 7] 7 For the Above Problem Statement, Describe the Approach using Sliding window technique and implement the solution in Java and analyze the time complexity.	5	1	4

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III- B. Tech. II Semester- KR23-MID-I Examinations, February/March - 2026

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Subject: **Cyber Security** | **PART-B** | Date: 25/2/2026-FN

Branch: CSE | Duration: 1Hr 40 Min. | Max. Marks: **20**

Answer any FOUR full questions from the following Questions **4 x 5 = 20 Marks**

Q.No	Questions	Marks	CO	BT
1.	An employee in the finance department receives an urgent email appearing to be from the company CEO. The email claims there is a "missed invoice" and demands the employee click a link to a "secure portal" to enter bank details immediately, or face legal action. Identify the specific type of attack and the vulnerability being exploited. Explain how this attack violates the CIA triad.	5M	CO1	3
2.a	Illustrate how packet analysis tools can be used to detect suspicious network activity.	2M	CO1	3
b.	Explain secure network protocols and describe how they ensure secure communication over a network.	3M	CO1	2
3.	Define assets, threats, vulnerabilities, and risks. Explain the risk management process in cybersecurity.	5M	CO1	3
4.	A feedback form displays user input without filtering special characters. **a)** Explain how Cross-Site Scripting (XSS) can be executed in this case. **b)** Discuss the risks posed to users. **c)** Apply sanitization and HTTPS implementation to mitigate the attack.	5M	CO2	2
5.	Explain broken authentication and session management vulnerabilities. Discuss secure authentication practices.	5M	CO2	2
6.	Describe Cross-Site Scripting (XSS) and its different types.	5M	CO2	3